Anti-GFP Tyramide Signal Amplification

Tyramide SuperBoost Kit, Alexa Fluor 488, Goat anti-Rabbit IgG (B40922) — adapted from User Guide MAN0015834 Rev. B.0

Volumes are for one 18 × 18 mm coverslip (about 100 µL per step). Scale for other sample sizes. Do not let samples dry out; use a humidified chamber for long incubations.

Reagents to prepare fresh on the day of use

Solution	Preparation
100X Tyramide stock	Dissolve Component C1 in 150 µL DMSO (Component E). Invert to dissolve. Stable 6 months at 2–8 °C.
100X H_2O_2 solution	1 drop (~50 µL) Component C2 in 1 mL distilled water. Prepare fresh.
1X Reaction buffer	1 drop (~50 µL) 20X Component C3 in 1 mL distilled water (Tris pH 7.4 also works). Prepare fresh.
Stop reagent stock	Add 1.45 mL 95% ethanol to Component D; vortex. Unused stock keeps at –20 °C for 6 months.
Stop reagent working soln.	Dilute stock 1:11 in PBS. Prepare fresh.

Procedure

1. **Fix and permeabilize.** To preserve native GFP, use the Image-iT Fixation/Permeabilization Kit (R37602). For tissue, deparaffinize and dehydrate first.
2. **Quench endogenous peroxidase (guide step 2.1).** Cover sample with undiluted 3% H_2O_2 (Component C2). Incubate **60 min**, room temperature.
3. **Rinse** 3 × with 1X PBS, room temperature.
4. **Block.** 100–150 µL Blocking buffer (Component A), 60 min, room temperature. (Biotin blocking not needed; no streptavidin is used.)
5. **Primary antibody.** Rabbit anti-GFP diluted in Component A. 60 min at room temperature or overnight at 2–8 °C. Use a much more dilute antibody than for standard IF (see optimization).
6. **Wash** 3 × 10 min in PBS.
7. **Poly-HRP goat anti-rabbit (Component B).** 100–150 µL, 60 min, room temperature. Shorten if non-specific signal appears.
8. **Wash** 3 × 10 min in PBS.
9. **Prepare tyramide working solution** (table below). Use within 2 h.
10. **Tyramide reaction.** Apply 100 µL; incubate 2–10 min at room temperature (start at 5 min).
11. **Stop.** Immediately add 100 µL Stop reagent working solution at the end of the timed incubation.
12. **Wash** 3 × with PBS.
13. **Counterstain** if desired (e.g. NucBlue, R37606).
14. **Mount** in ProLong Diamond (P36961) or SlowFade Diamond (S36963). Image with a standard 488/FITC filter set.

Tyramide working solution	1 coverslip (~100 µL)	5 coverslips
100X Tyramide stock	1 µL	5 µL
100X H_2O_2 solution	1 µL	5 µL
1X Reaction buffer	100 µL	500 µL

Optimization (first run)

- **Primary dilution:** your standard dilution, 5-fold more dilute, and 50-fold more dilute, plus a no-primary negative control.
- **Tyramide time:** test 0, 2.5, 5, 7.5 and 10 min with positive and negative controls.
- **H_2O_2 quench:** optional for most cultured cells; compare a sample with and without it. More useful for tissue.

Troubleshooting

Observation	Action
Excess signal	Dilute primary further; shorten tyramide time; lower tyramide concentration
Low signal	Optimize primary dilution/time; lengthen tyramide time; use antigen retrieval
Blurry / low resolution	Shorten tyramide time; check stop reagent dilution
High background	Lengthen H_2O_2 quench; dilute primary; longer block; more/longer washes; shorten tyramide or secondary step; lower secondary concentration

Note: Alexa 488 tyramide overlaps native GFP. Use a different tyramide (e.g. Alexa 594 or 647) if you want to distinguish amplified from native signal. DMSO is hazardous; avoid skin and eye contact.